

MAE 3127

Fluid Mechanics Laboratory



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Topics covered today:

- Impact of Jet on a Surface – Theoretical estimation of reaction force to hold the surface stationary



School of Engineering
& Applied Science

THE GEORGE WASHINGTON UNIVERSITY

Fall 2025

Photo: Kartik Bulusu



“Study hard what interests you the most in the most undisciplined, irreverent and original manner possible.”

— Richard Feynmann

For example,

Reaction force to hold a surface, stationary under the impact of a jet - a theoretical estimate



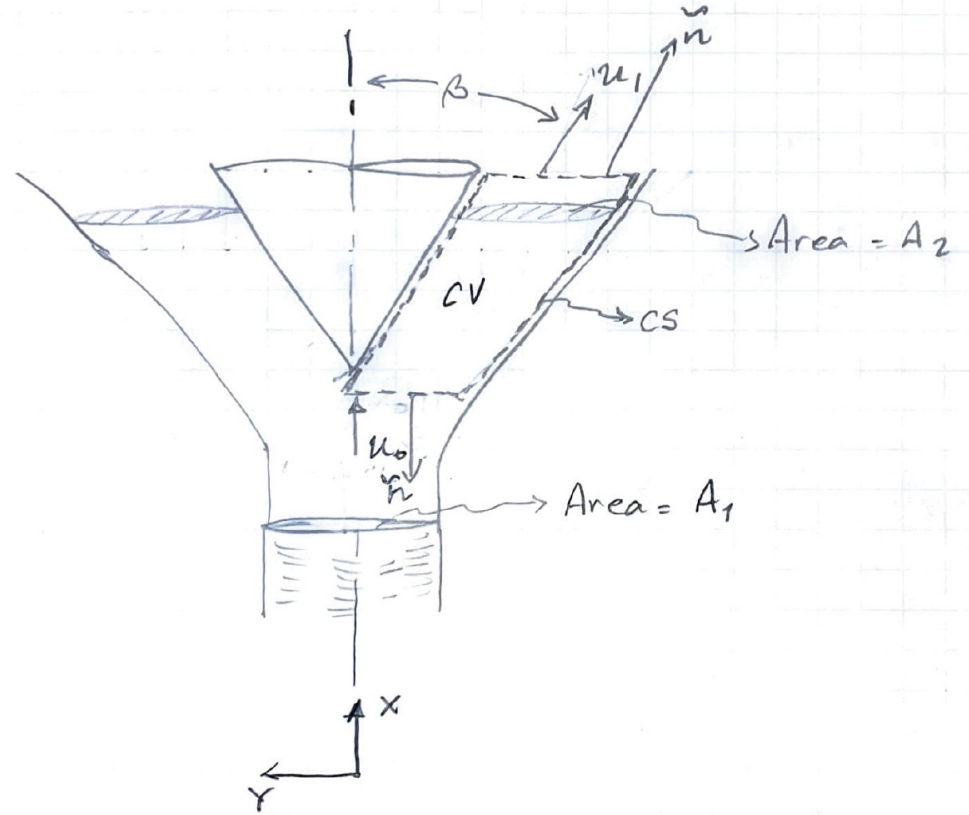
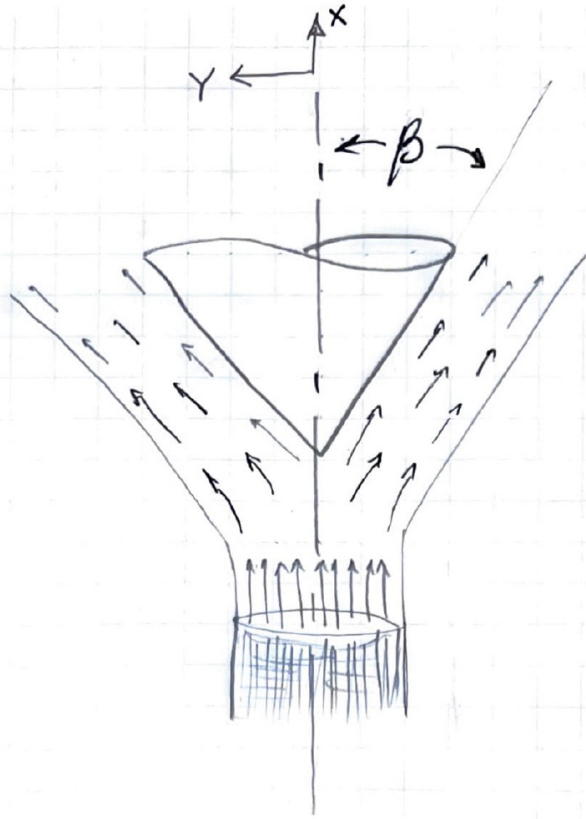
Points to ponder on after you leave the recitation today:

- Take all the notes you want, ask all the questions you want.
- You are encouraged to study in groups, but you may **not** show HWs among yourselves.
- Under **no circumstances** may you look at another student's HWs, or look for HW answers anywhere other than in the text and in the notes and handouts provided in class or links provided on the course website. If you do, it is considered an academic integrity violation!
- **Collaborate but don't violate!**

On deliverables:

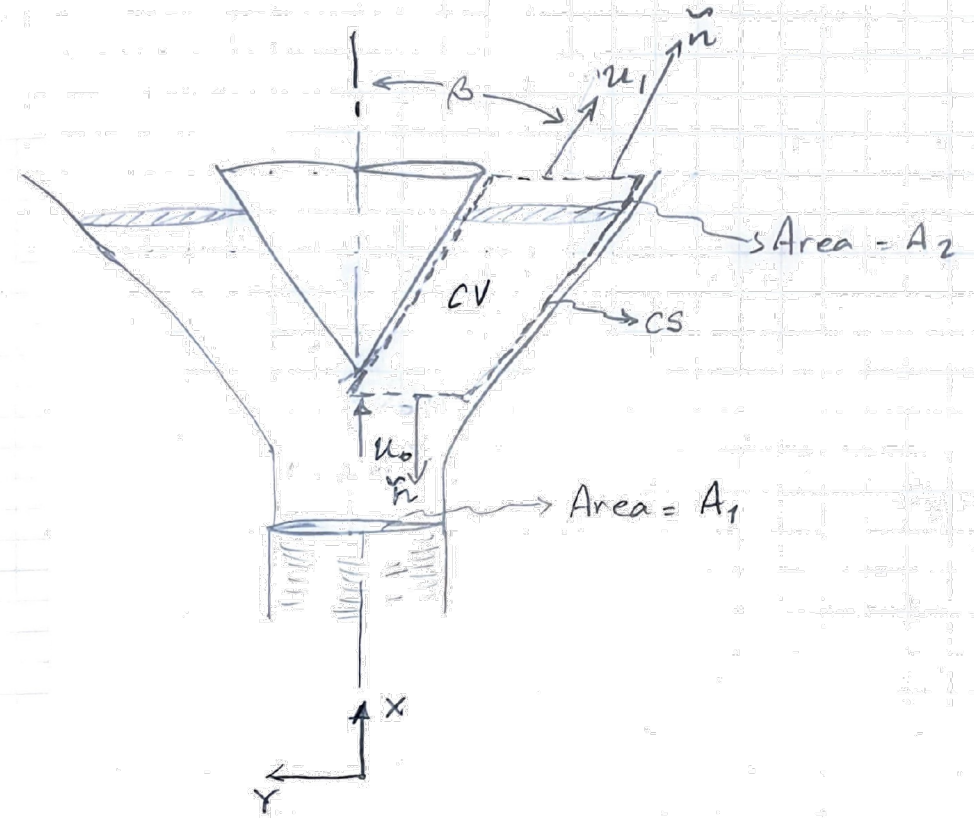
- **Deadline for submitting the Lab deliverables** is 10/22.
- You need to **upload all deliverables as a Zip-file**, on Blackboard.
- **Please write clearly and legibly and upload clear images** to be graded.
- Utilize office hours.
- Check Ed discussion board: <https://edstem.org/us/courses/83318/discussion>





Assumptions:

- ① Steady flow
- ② Incompressible flow
- ③ Neglect friction between the impact surface and jet
- ④ Magnitude of the jet velocity does not change as the jet turns. $\Rightarrow |u_0| = |u_1|$
- ⑤ Uniform velocity profiles at the control surfaces
- ⑥ Jet is circumferentially symmetrical
- ⑦ Neglect body forces



Conservation of Mass:

$$-\frac{\partial}{\partial t} \iiint_{CV} \rho dV = \oint_{CS} \rho (\vec{v} \cdot d\vec{A})$$

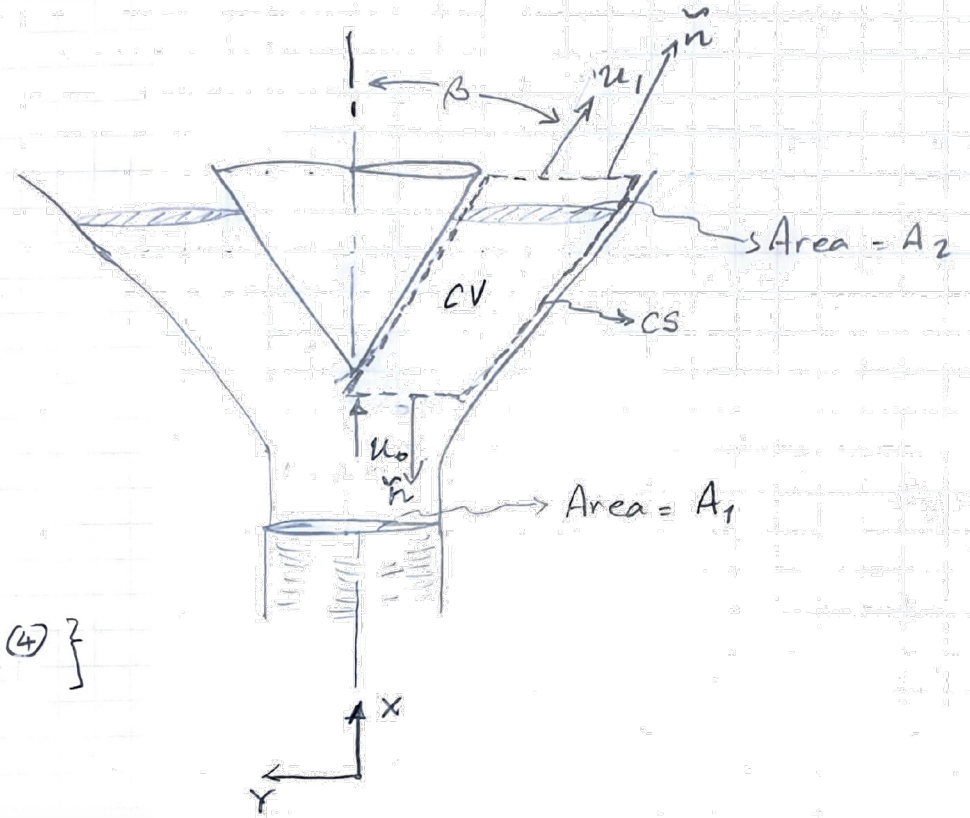
$$\Rightarrow -\iiint_{CV} \frac{\partial}{\partial t} (\rho dV) = \oint_{CS} \rho (\vec{v} \cdot d\vec{A})$$

0, steady flow

$$\Rightarrow \oint_{CS} \rho (\vec{v} \cdot d\vec{A}) = 0 \rightarrow \textcircled{1}$$

$$\rho(-u_0)A_1 + \rho(u_1)A_2 = 0 \quad \left\{ \because \text{From Assumption (4)} \right. \\ \left. |u_0| = |u_1| \right\}$$

$$\Rightarrow -A_1 + A_2 = 0 \Rightarrow \boxed{A_1 = A_2} \rightarrow \textcircled{2} \\ (\text{no spreading!})$$



Conservation of Linear momentum :

$$\vec{F} = \oint_{CS} \vec{x} (\rho \vec{v} \cdot d\vec{A}) + \frac{\partial}{\partial t} \left(\iiint_{CV} \vec{v} (\rho dv) \right)$$

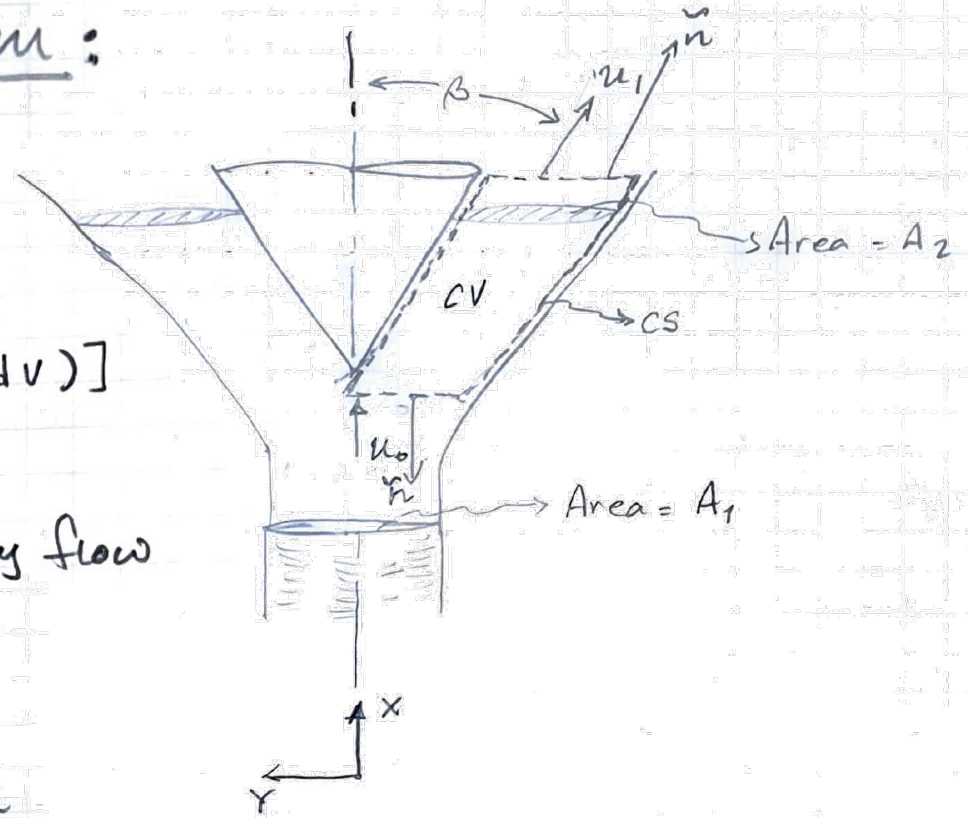
$$= \oint_{CS} \vec{v} (\rho \vec{v} \cdot d\vec{A}) + \iiint_{CV} \frac{\partial}{\partial t} [\vec{v} (\rho dv)]$$

0, steady flow

$$\Rightarrow \vec{F} = \oint_{CS} \vec{v} (\rho \vec{v} \cdot d\vec{A}) \rightarrow \textcircled{3}$$

External
force
Resultant

Rate of momentum
efflux across
control surface



In the direction of the jet i.e., X-direction we have

$$\Rightarrow F_x = (-u_0)(\rho u_0 A_1) + (u)[\rho(u_0 \cos \beta) A_2]$$

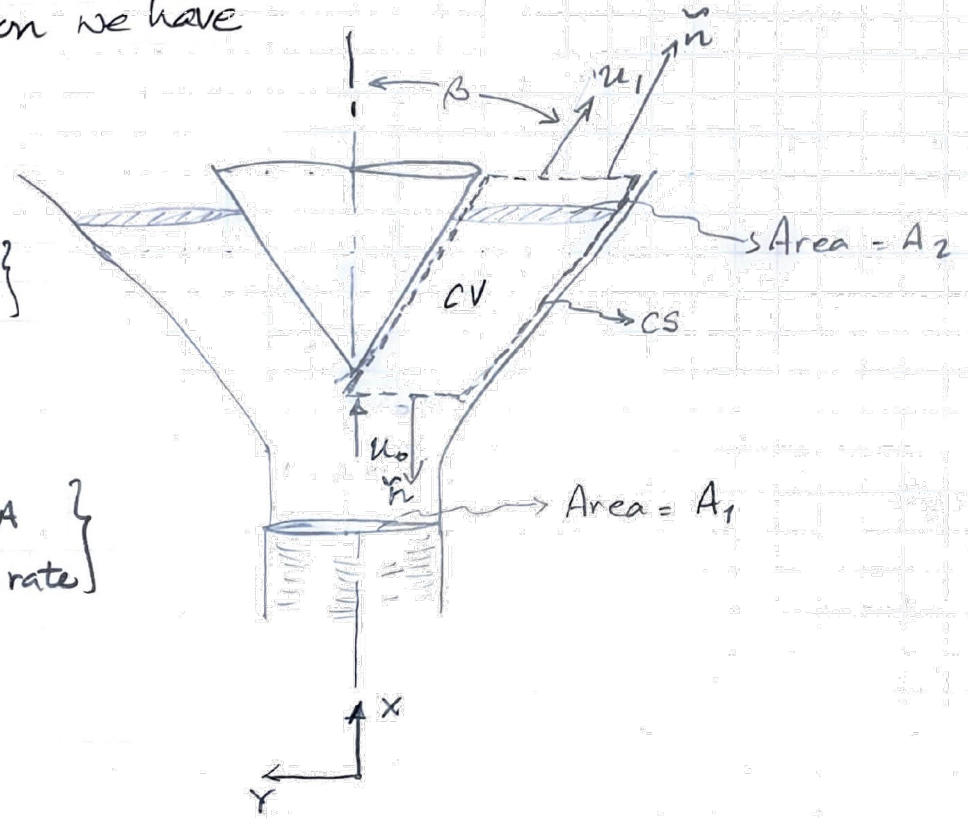
From (2), $A_1 = A_2$ & $\left\{ \begin{array}{l} \text{From assumption (4)} \\ |u_0| = |u_1| \end{array} \right\}$

$$\Rightarrow F_x = -u_0^2 \rho A + u_0^2 \rho \cos \beta A$$

$$\Rightarrow F_x = (-u_0)(\rho u_0 A) + (u_0 \cos \beta)(\rho u_0 A) \quad \left\{ \begin{array}{l} \therefore \dot{m} = \rho u A \\ \text{or} \\ \text{mass flow rate} \end{array} \right\}$$

$$\Rightarrow F_x = -\dot{m} u_0 + \dot{m} u_0 \cos \beta$$

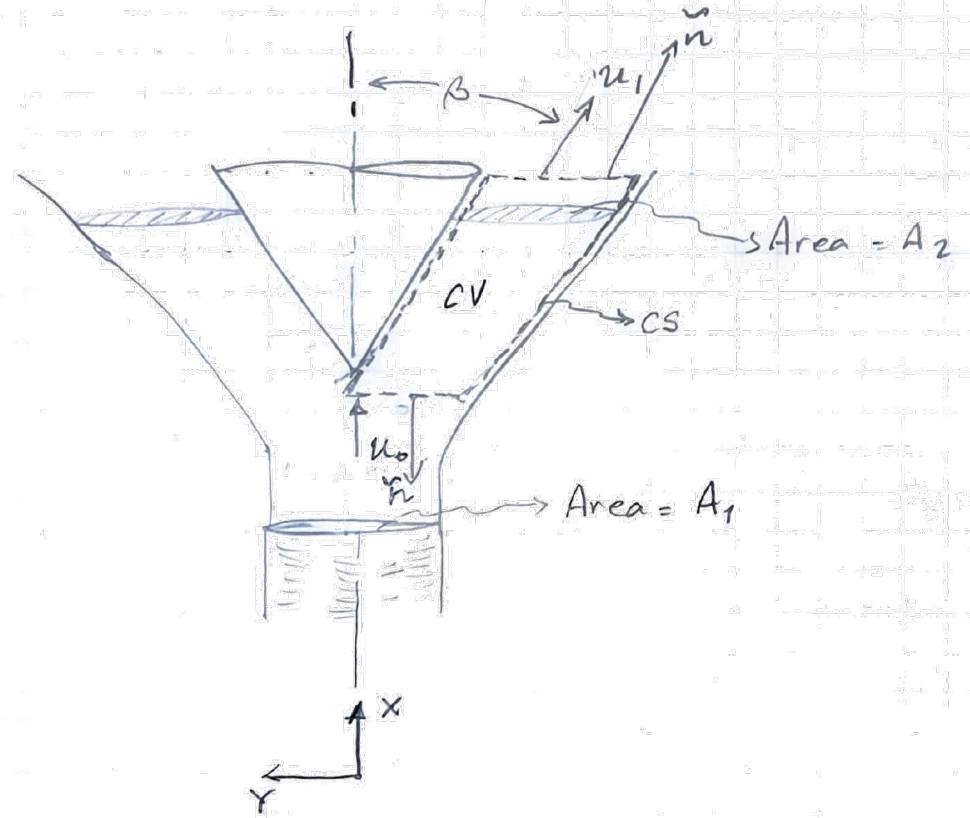
$$\Rightarrow \boxed{F_x = \dot{m} u_0 (-1 + \cos \beta)} \rightarrow (4)$$

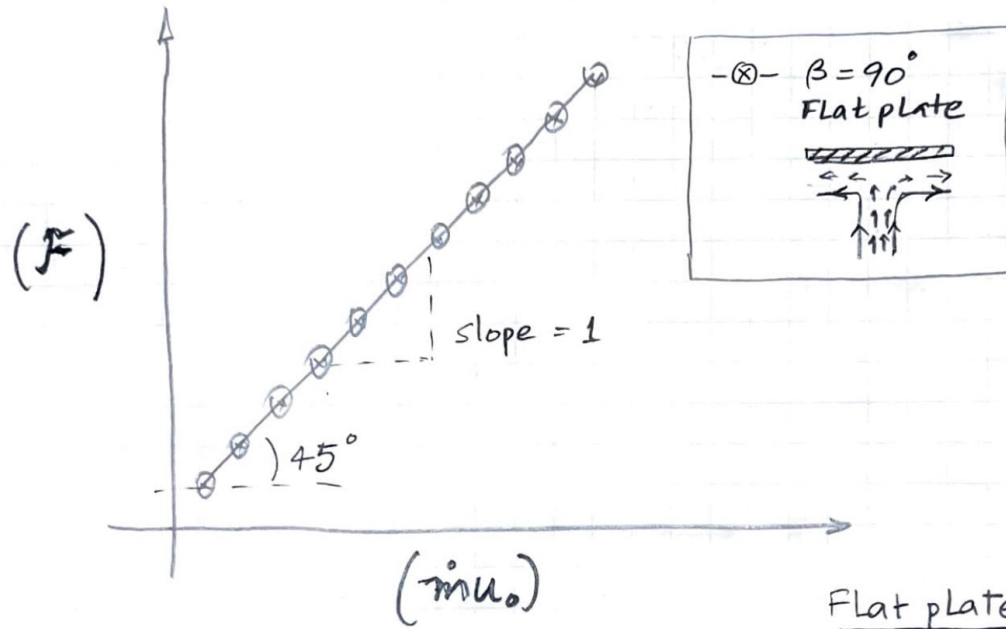


Reaction force in X-direction
necessary to hold the impacted
surface stationary:

$$\boxed{F = -F_x} \rightarrow (5)$$

$$\Rightarrow \boxed{F = \dot{m} u_0 (1 - \cos\beta)}$$





$$F = \dot{m}u_0(1 - \cos\beta)$$

Flat plate

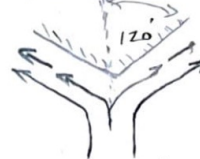


$$\beta = 90^\circ$$

$$\cos\beta = 0$$

$$(1 - \cos\beta) = 1$$

120° Cone



$$\beta = 120^\circ$$

$$\cos\beta = -0.5$$

$$(1 - \cos\beta) = 1.5$$

hemisphere



$$\beta = 180^\circ$$

$$\cos\beta = -1$$

$$(1 - \cos\beta) = 2$$

30° Cone



$$\beta = 30^\circ$$

$$\cos\beta = 0.87$$

$$(1 - \cos\beta) = 0.13$$

